

# Forensics for Cybercrimes Committed Using Artificial Intelligence

A Digital Forensic Framework for AI-Assisted Cybercrime

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# Research Context

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Artificial Intelligence is changing the cybercrime landscape.

## Traditional Cybercrime

- ▶ Human-controlled systems
- ▶ Identifiable devices
- ▶ Conventional malware
- ▶ Established forensic procedures

## AI-Assisted Cybercrime

- ▶ Automated attacks
- ▶ Synthetic identities
- ▶ AI-generated content
- ▶ Autonomous decision-making

# The Forensic Challenge



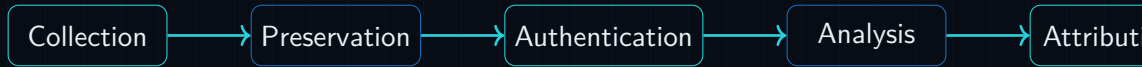
The central question is not merely:

*What happened?*

but also:

*Who or what caused it, and can we prove it?*

# Forensic Evidence Pipeline



# Formalising the Evidence Problem

Let

$$E = \{e_1, e_2, \dots, e_n\}$$

represent the set of collected digital evidence.

Let

$$A \subseteq E$$

represent evidence that has successfully passed an authentication process.

Therefore,

$$A \subseteq E$$

and

$e_i \in A \Rightarrow e_i$  satisfies the defined authentication criteria.

# Research Objectives

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1. Examine how AI can facilitate cybercrime.
2. Identify limitations in existing digital forensic approaches.
3. Investigate evidence provenance and integrity.
4. Develop a forensic framework for AI-assisted cybercrime.
5. Evaluate the proposed framework.

## Expected Contribution



# Conclusion

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## Central Argument

AI changes not only how cybercrimes can be committed, but also how digital evidence must be interpreted, authenticated and attributed.

*AI + Cybercrime + Digital Forensics + Evidence*



# Thank You

## Questions & Discussion

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